

Aryan Singh

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EDUCATION

Johns Hopkins University

Baltimore, MD

B.S. Computer Science & Biomedical Engineering | GPA: 4.00

Aug 2024 – May 2028

- Minor: Entrepreneurship & Management
- **Coursework:** Probability & Stochastic Processes, Algorithms, Linear Algebra, Numerical Methods, Optimization, Statistical Inference, Systems Programming (C/C++), Quantum Computing, Discrete Mathematics

QUANTITATIVE PROJECTS

IMST-Quant: GNN Sentiment Trading Engine | *Python, PyTorch Geometric, FinBERT, Spark*

- Built an end-to-end trading system with GNN author-influence scoring (PyTorch Geometric), FinBERT sentiment, and a 6-layer medallion architecture generating alpha from Reddit, equity, and crypto data under strict T-1 causality to eliminate lookahead leakage
- Benchmarked LSTM, CNN, and Transformer forecasters via event-driven backtesting with transaction costs, slippage, and walk-forward optimization across 5+ years of market data
- Built risk analytics computing Sharpe, max drawdown, VaR, and rolling beta, with Kelly-criterion position sizing and dynamic stop-loss optimization; developed in unit-tested, code-reviewed modules

IMC Prosperity 4: Algorithmic Trading Competition (Top 500) | *Python, NumPy, pandas*

- Built market-making, statistical-arbitrage, options-pricing, and cross-asset-arbitrage strategies optimized for sub-900ms latency with dynamic position limits and persistent state across rounds
- Implemented implied-volatility estimation, mean-reversion detection, and order-book imbalance scoring across a live options and equities microstructure

EXPERIENCE

LabLink – AI-Powered Research Platform

Jan 2025 – Present

Co-Founder & Lead Engineer

Baltimore, MD

- Built and shipped an AI research platform (published on npm) with LLM-integrated NLP pipelines and automated workflow tooling, using AI agents to accelerate development and drive engineering efficiency across the team

NeuroVoice – AI Diagnostics Startup

Mar 2024 – Aug 2025

Machine Learning Engineer, Founding Team

Houston, TX

- Architected a multimodal inference pipeline (PyTorch) with Bayesian model selection, Optuna hyperparameter optimization, and rigorous evaluation (ROC-AUC, calibration curves, bootstrap CIs) across 12TB+ of data, delivering 25–30% accuracy gains
- Cut per-inference cost from \$0.12 to \$0.003 (97.5%) via model distillation, INT8 quantization, and infrastructure optimization; built a cost-performance Pareto frontier analysis to drive deployment decisions
- Designed a real-time drift-detection and A/B testing framework monitoring model degradation under distribution shift, seamlessly reconciling historical and streaming data

Texas Center for Superconductivity, University of Houston

Feb 2023 – Mar 2024

Quantitative Research Engineer

Houston, TX

- Built quantitative pipelines (PCA, t-SNE, ResNet-50, EfficientNet) over 120,000+ samples at 94%+ accuracy, and a constrained-optimization framework for Pareto-optimal parameter search that cut experimental cycles by 80%

Rice University

Sep 2021 – Dec 2022

Computational Genomics & Statistical Modeling Researcher

Houston, TX

- Applied statistical modeling (logistic regression, XGBoost, CNN-LSTM) to 10+ TB of data with stratified cross-validation and calibrated probability estimation, reducing false negatives 60%
- Engineered parallelized pipelines (Snakemake, SLURM) achieving 40x throughput on terabyte-scale datasets with automated QC and reproducible environments

HONORS & AWARDS

USACO Platinum (top 0.25%) | **USAMO Qualifier** | **AIME 4x Qualifier** (top 5% of 300K+) | **USABO Top 100** | **USNCO Nationals** | **USAPhO Silver Medal** | **Rice Neurotransmitter Research Conference Grand Prize**

TECHNICAL SKILLS

Languages: Python, C++, C, Java, SQL, MATLAB, R, Bash

Quantitative: Stochastic calculus, time-series analysis, Monte Carlo simulation, statistical arbitrage, portfolio optimization, risk metrics (VaR, Sharpe, drawdown), Bayesian inference, options pricing, signal processing

ML & Data: PyTorch, scikit-learn, XGBoost, PyTorch Geometric, pandas, NumPy, SciPy, statsmodels, Spark

Infrastructure: Docker, AWS, Git, Linux, CI/CD, unit testing, code review, PostgreSQL, Airflow, low-latency systems, distributed computing